CSCI 513.01W – Python Programming for AI
Assignment #7

Spring 2025

Due Date: Sunday April 20th

Total Points: 100

Suppose we have a company that purchases cars online. We want to automate the process of evaluating a car in order to determine whether to purchase it or not. Let us develop three different types of classifiers by answering the questions below based on the dataset called **cars.csv**. Each row in the dataset describes specific car properties with a class label that indicate whether we buy the car or not.

Questions

For each question below, **only** use Pandas, Plotly, and Python to write one or more instructions to compute the answer. Use one Notebook cell for each question. The **only** Pandas, Plotly, and Python features that you can use in this assignment are the ones introduced in class.

1. How many cars does this dataset have?
2. How many cars in this dataset were bought?
3. How many cars in this dataset have a very high price?
4. How many of the very high-priced cars were purchased?
5. Using Plotly, visualize the relationship between car prices and whether the car is sold or not.
6. Using Plotly, and from the cars that were only sold, use a bar diagram to visualize the relationship between the requested price and maintenance price.
7. Using Plotly, and from the cars that were only sold, visualize the relationship between car safety and maintenance price.
8. Find any other interesting relationship in the data, and visualize it using Plotly.
9. Create two variables called X and y, where y stores all the class labels, while X stores all the class features not including the class label.
10. Next, we want to convert the data in X from string to integer. We are going to use a library called *category_encoders* to do that. First, let us install this library:
    pip install category_encoders

    Then, run the following code:
    ```
    from category_encoders import OrdinalEncoder
    encoder = OrdinalEncoder()
    X = encoder.fit_transform(X)
    ```
11. Divide the dataset into training and test sets. The training set should be 70% while the test set should 30%. Use 53 as the random state. How many rows does the training set have, and how many rows does the testing set have?
12. Create a logistic regression classifier, then train and test your classifier. What is the accuracy, precision, recall and f1-score of the model on the test data?

13. What are the true positives, true negatives, false positives, and false negatives?
14. Although we did not learn about Support Vector Machine (SVM), let us use this classifier. Different from logistic regression that only uses linear equation, SVM supports different types of equations, including linear equation. These equations in SVM are called kernels. RBF (Radial Basis Function) is one of the kernels that is a popular in classification. RBF requires one parameter called *gamma* that we need to specify in Sklearn. Gamma should have a value from 0 to 1, where a higher value will cause the SVM to fit (memorize) the training dataset, which we do not want to do. Usually, a gamma value of 0.1 is usually used. The value of gamma needs to be manually specified in the learning algorithm. Create an SVM model with an RBF kernel. Then, train and test your classifier. What is the accuracy, precision, recall, f1-score and confusion matrix of the model?
15. Similar to SVMs, we also did not learn about random forest as a classifier but let us try it out. Create a random forest model with 100 trees, the number of trees can be specified as a parameter in Sklearn. Then, train and test your classifier. What is the accuracy, precision, recall, f1-score and confusion matrix of the model?

What to hand in

Submit your program electronically through D2L. Please hand in your Jupyter notebook file (Python program: .ipynb)